2021 F=ma Exam: Problem 5

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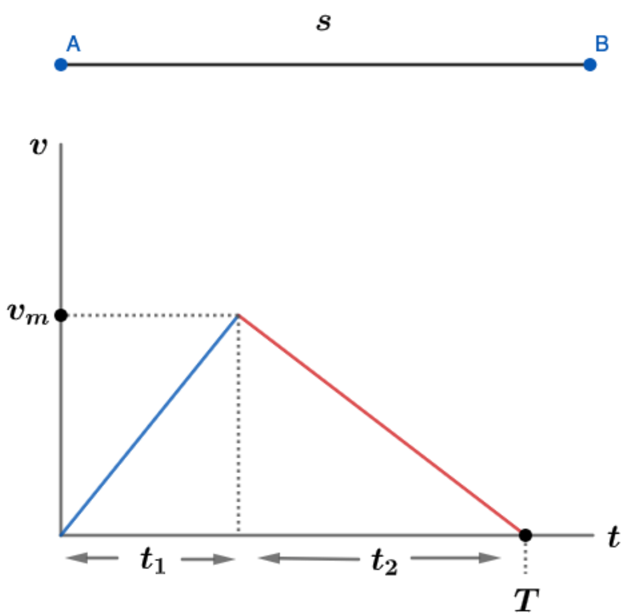

For the path of shortest time T, the train can travel a maximum distance s. By the v-t graph, we see that to maximize the displacement (area under the curve), the train will accelerate at a_1 to some maximal velocity v_m then decelerate at a_2 to rest. From kinematics,

$$v_m = a_1 t_1 = a_2 t_2$$

$$s = \frac{1}{2} v_m T$$

Since $t_1 + t_2 = T$, we can substitute from the above equations,

$$\frac{v_m}{a_1} + \frac{v_m}{a_2} = \frac{2s}{T}\left(\frac{1}{a_1} + \frac{1}{a_2}\right) = T$$

$$T^2 = 2s\frac{a_1 + a_2}{a_1 a_2}$$

$$T = \sqrt{\frac{2s(a_1 + a_2)}{a_1 a_2}}$$

Thus, the answer is $\boxed{\text{C}}$.